POLITECNICO MILANO 1863 | DIPARTIMENTO DI SCIENZE E TECNOLOGIE AEROSPAZIALI

Combustion in Thermochemical Propulsion: Assignments (A1, v01b)

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Abstract

The presentation introduces the assignments for the current academic year. Two assignments are given: report A1 must be taken by both 8 and 6 CFU students, report A2 is for students of the 8 CFU track only.

A background is given for both the assignments, and reequirements of each problem are clarified.

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1. Assignement A1: Analysis of the Thermal Behavior of Thermites – THREAD
    a. Background
    b. Toward Thermite for Demise: EIC THREAD
    c. Fundamentals of Thermal Analysis
    d. Raw Data

2. Assignment A2: Preliminary Development of Solid Propellent Formulations for Distress Signals
    a. Background
    b. Requirements

Assignment A1

01

Objective:

Provide kinetic parameters and reactivity of thermite formulations

working on isoconversional methods

Il concetto di utilizzare la termite per la "demise" (ovvero la distruzione intenzionale o lo smaltimento rapido di hardware, dati o componenti strutturali) si basa su una reazione redox (ossido-riduzione) estremamente esotermica

Students will achieve knowledge of

i. Fundamentals of thermite reaction (introduction to pyretechnic formulations)
ii. Non-conventional application of thermite formulations: thermite for demise concept
iii. Differences in the reactivity of parameters when the same thermite is prepared/crafted in different ways/systems

Why is this relevant?

i. Thermite for demise (T4D) show the relevance of energetic materials for safer, more sustainable and efficient re-entry of satellites in low-Earth orbit (LEO)
ii. Using thermites for satellite demise requires a detailed understanding of thermite reaction process, and a fine tuning of thermite reactivity

A1: Background (1 of 5)

in the products of this pyrotechnic process there are a lot of condensed species (this is a huge difference with respect propellants, that generate gasseous species mainly as products)

In the field of energetic materials, *thermite* to a pyrotechnic composition based on solid-state reactants governed by a specific reduction-oxidation (redox) path. The most comprehensive and clear definition for the classification of thermitic reactions and compositions is provided by Shaw [1] (see the hereby reported extract), while Fischer and Grubelich [2] and Weiser et al. [3] give a detailed catalogue of theoretical energy release from several thermites.

> *[...] these are reactions in which an electronegative atom is transferred from one electropositive atom to another electropositive atom. The electronegative element is often oxygen, although it does not need to be. The two electropositive elements are often metals, but they may also be metalloids or, less frequently, nonmetals [...].*

M = metal

you can see the passage of the oxygen from one metal to the other

$$M^1 + \frac{a}{b \cdot c} M_d^2 O_b \rightarrow \frac{1}{c} M_c^1 O_a + \frac{d \cdot a}{b \cdot c} M^2 \quad \Longrightarrow \quad Al + \frac{1}{2} Fe_2O_3 \rightarrow \frac{1}{2} Al_2O_3 + Fe$$

[1] Shaw A.P.G., *Thermitic Thermodynamics - A Computational Survey and Comprehensive Interpretation of Over 800 Combinations of Metals, Metalloids, and Oxides*, CRC Press, 2020, https://doi.org/10.1201/9781351056625.
[2] Fischer, S. H. and Grubelich, N. C., *Theoretical Energy Release of Thermites, Intermetallics, and Combustible Metals*, in: 24th International Pyrotechnics Seminar, 1998, https://doi.org/10.2172/658208.
[3] Weiser, V. Roth, E. et al., Experimental and Theoretical Comparison of a SystematicVariety of Thermite Compositions, accessible at: publica-rest.fraunhofer.de/server/api/core/bitstreams/fa6230b1-4cec-4b7f-8700-6712efdc67df/content

A1: Background (2 of 5)

Why are thermites so relevant?

ignitability = how much energy you have to deliver to the system in order to get combustion conditions --> IGNITION

- **Energy density**: thermites offer a higher energy density than conventional explosives [4, 5].
  - On the other hand, the released thermal power is lower for thermites than for explosives, due to the longer reaction time of the former. the reactions in thermites is slower than the one occuring in explosives
    - Thermite reactions → diffusion phenomena, whereas explosives → chemical kinetics.
- **Tailorable ignitability**: driving parameters for the ignitability and the reactivity of thermites are the particle size and morphology of the ingredients of the composition. and how the ingredients are shaped / how the charges are connected
  - Aluminothermic reaction has been used for years as a welding solution when working in isolated areas (e.g., railway construction)
- **Tunable reactivity**: particle size (micron- vs. nano-sized), particle morphology, and production method (e.g., loose separated powders, pressed pellets, mechanically activated powders) influence the propagation speed of the thermite. reactivity: how fast the reaction is propagating ---> PROPAGATION

General, if something ignites fast also propagates fast, but it is not always true

[4] Granier J.J. and Pantoya M.L., *Laser ignition of Nanocomposite Thermites*, Combustion and Flame, Vol. 138, No. 4, 2004, doi: https://10.1016/j.combustflame.2004.05.006.
[5] Fahd, A. et al., *Combustion characteristics of EMOFs/oxygenated salts novel thermite for green energetic applications*, Thermochimica Acta, Vol. 704, 179019, doi: https://10.1016/j.tca.2021.179019.

Il processo di attivazione tramite ball milling (macinazione a sfere) è una tecnica di ingegneria dei materiali utilizzata per modificare le proprietà fisico-chimiche di una sostanza attraverso sollecitazioni meccaniche ad alta energia.

Il processo avviene all'interno di un contenitore rotante o vibrante caricato con sfere pesanti (di acciaio, ceramica o carburo di tungsteno) e la polvere da trattare. Collisioni ad alta energia: Le sfere colpiscono le particelle di polvere contro le pareti del contenitore e tra di loro.

A1: Background (3 of 5)

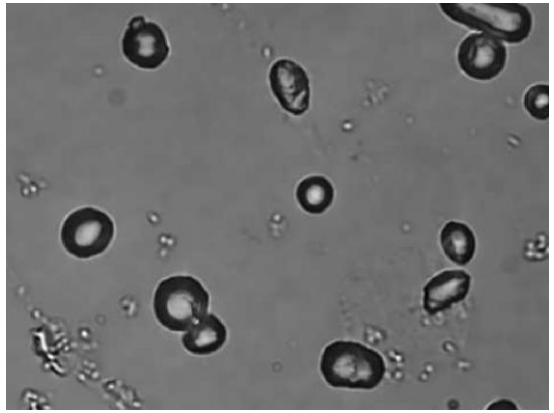
How are thermites classified? based on particles size

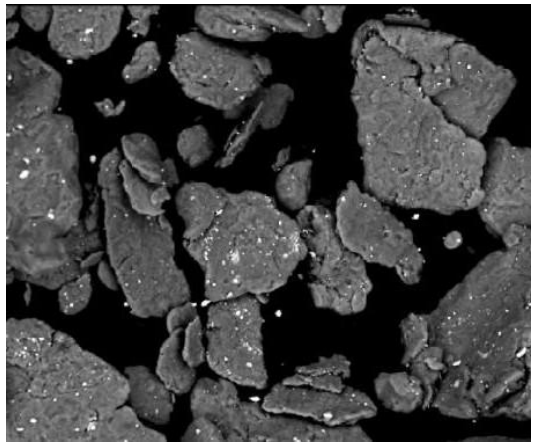
lower reactivity (initial threshold is high)

- **Conventional Thermites**: thermites based on sub-millimeter or micron-sized ingredients.
  - Conventional thermites can be based on advanced ingredients as *activated powders*. These are disperse systems whose features are altered by mechanical, chemical or mechano-chemical processes. Mechanically activated powders are produced starting from regular ingredients (spherical particles) whose shape and (possibly) composition can be modified by the mechanical stresses produced during low-/high-energy processes.

or chemical modifications

mechanical activation process --> ball milling: take the powder (that can be made with different ingredients), put it in a jar, then rotation. Thus, inside the jar, a lot of stresse are present and they will alter the shape of the particles





in this way, you can REDUCE THE DIFFUSION DISTANCE; also, layering becomes possible

A1: Background (3 of 5)

How are thermites classified?

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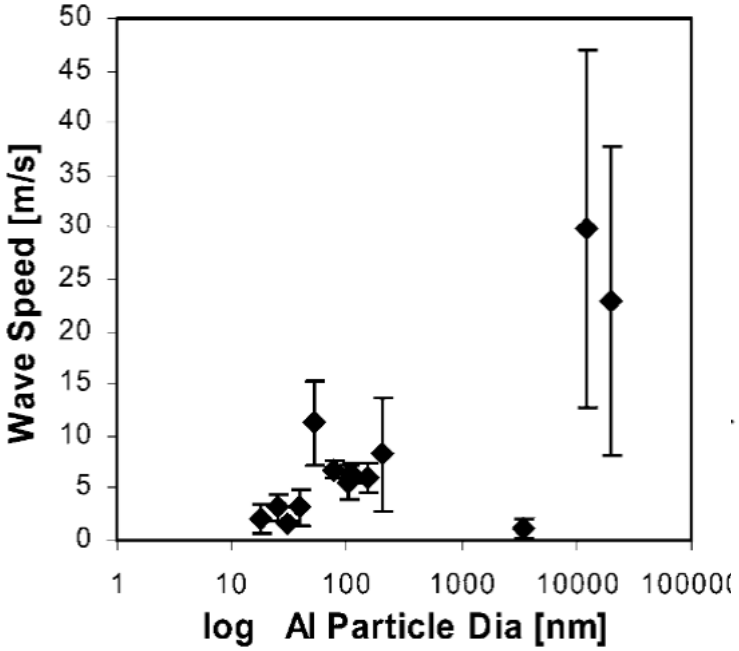
- **Nano-sized Thermites (aka metastable intermolecular composites – MICs):** these formulations are based on nano-sized ingredients. While also nano-sized thermites can undergo activation processes, this is typically avoided thanks to their intrinsically high reactivity (reduced size → increased surface area).

specific surface area SSA = (4*pi*R^2) / (4/3 * pi * R^3 * rho) = surface / mass that is approximately 1/R

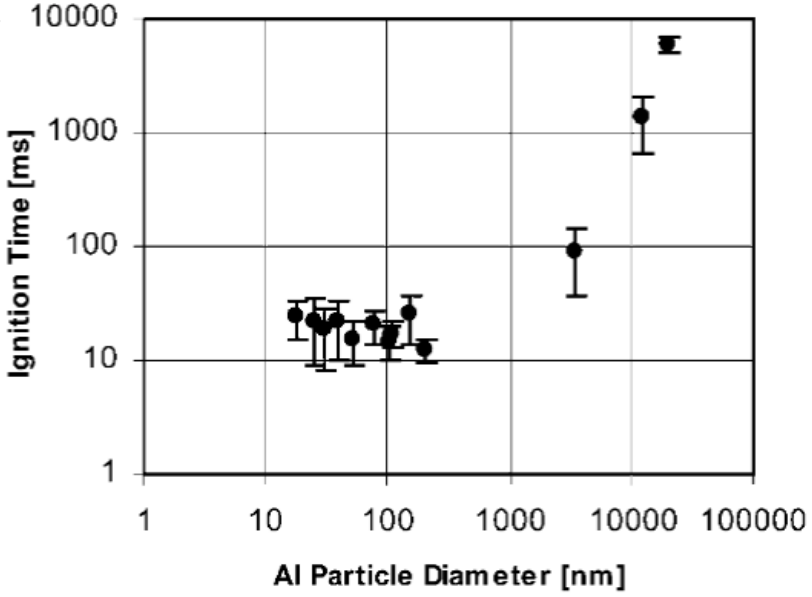
THE LARGER THE PARTICLE , THE SMALLER THE SSA.

SSA acts as the window through which the particle senses the world, thus, reducing SSA means easier ignition and faster combustion.

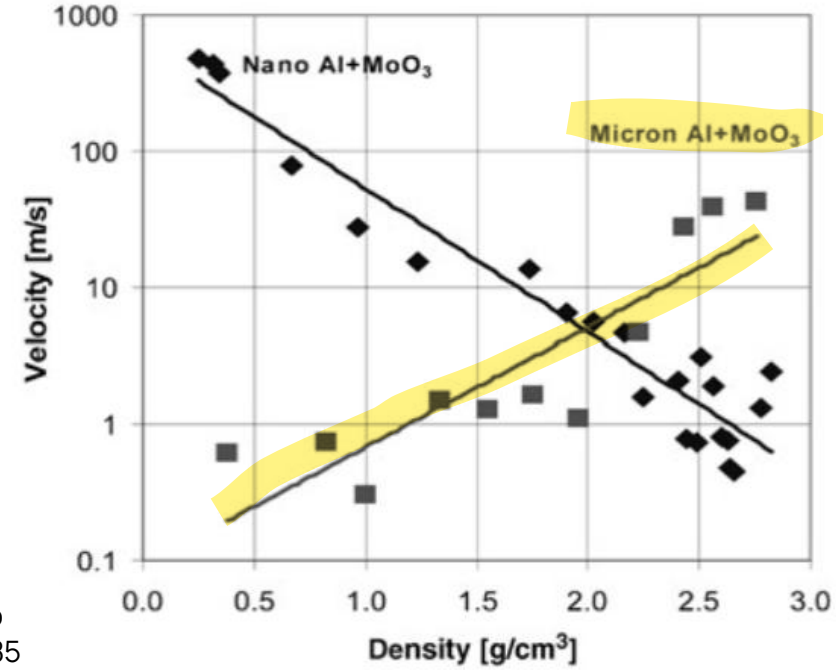
A1: Background (4 of 5)



reduction in the ignition time is achieved reducing the particle's diameter.



micron: increasing the density, you are reducing the diffusion distances and thus increasing the velocity



Images are taken from: Pantoya, M. and Granier, J. (2005), Combustion Behavior of Highly Energetic Thermites: Nano versus Micron Composites. Propellants, Explosives, Pyrotechnics, 30: 53-62. https://doi.org/10.1002/prep.200400085

A1: Background (5 of 5)

What is design for demise (D4D)? How can this be translated into thermite for demise (T4D)?

some s/c with thermochemical propuslion (= tanks) or reaction wheels are bulky ssystems and thus difficult to be burnt by the atmosphere only

- D4D is a design methodology targeting the full ablation of a space object during re-entry. The objective is reducing the casualty risk of an uncontrolled re-entry below 1 in 10000.
- Energetic materials can be used to increase/tailor the heat load during the re-entry → T4D
- Thermites feature suitable characteristics (reduced production of gaseous species with respect to propellants, more tailorable than explosives)
- The concept of T4D is originally proposed by Dihlan and Omaly (*Patent FR 2975080B1*, 2011), Seiler and Smet (European Pat. 3604143A1) Monogarov et al. https://doi.org/10.1016/j.actaastro.2016.10.031, https://doi.org/10.1016/j.actaastro.2020.03.028, and is further developed in SPADEXO and THREAD projects funded by ESA and EU, respectively.
- Several publications are available in the frame of the SPADEXO project (https://doi.org/10.1016/j.actaastro.2024.06.013, https://doi.org/10.1007/s12567-025-00598-8)

benefit: few components that can fail

EIC THREAD

THermite REactions Assisting satellite Demise

EIC Pathfinder Open project

Full project details are available at Thread - T4D, in particular, the project's aims and the composition of the consortium are presented.

Here we focus on the WP2 of the project.

WP2 working team (current, alphabetical order): Federica Catani, Valentina D'Annunzio, Jacopo Domaschio, Filippo Maggi, Christian Paravan, Alberto Verga, and Carlo Zanardi.

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Grant Agreement 101186901

European Innovation Council

EIC THREAD

THermite REactions Assisting satellite Demise

EIC Pathfinder Open project

CURRENT RESEARCH TOPIC

Is it possible to shape thermite charges capable of maintaining their shape and/or bear a structural load throughout a spacecraft's mission, and deliver heat precisely to assist the demise of said spacecraft?

'Building Blocks (BB)': thermite elements with structural integrity

Slide credit: Jacopo Domaschio

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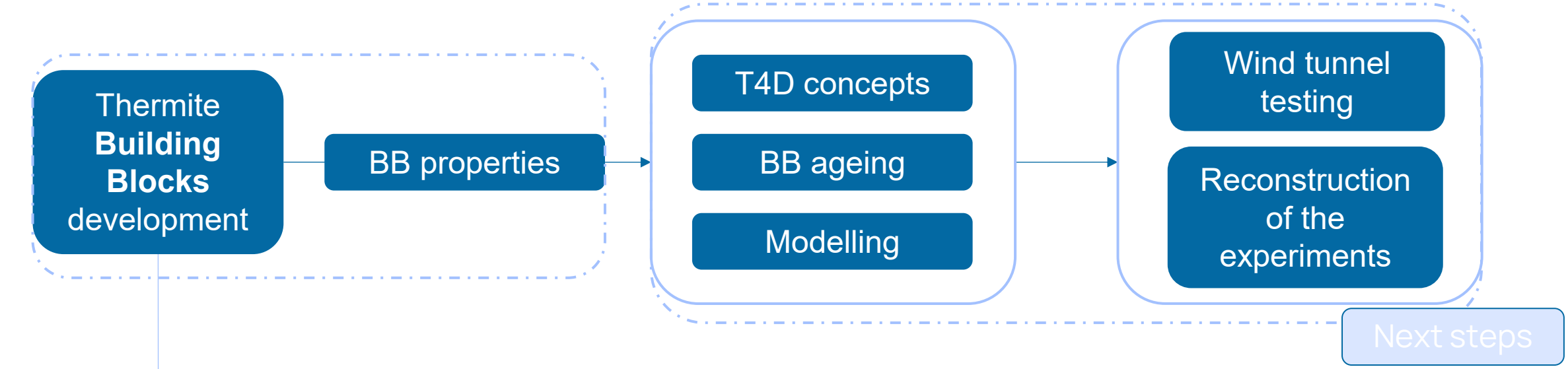
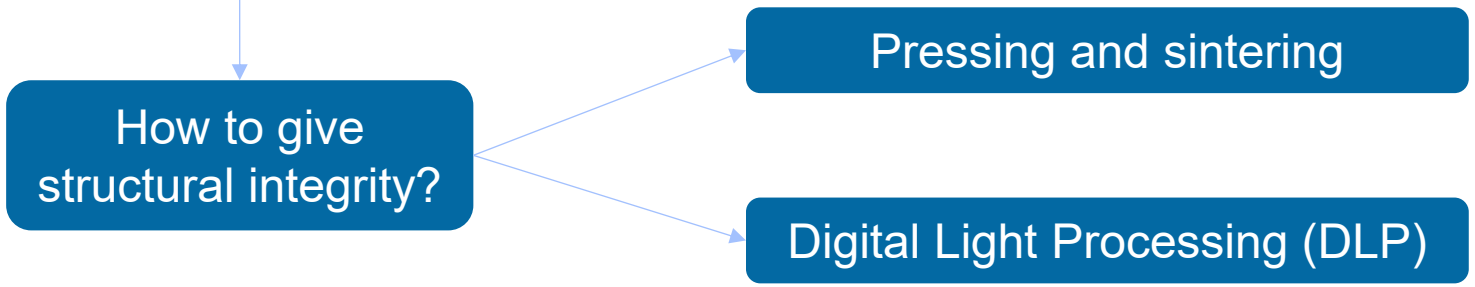
EIC THREAD

Slide credit: Jacopo Domaschio





Exploit metal fuel to gain structural properties via **sintering**

Embed thermite powder into **UV-curing resin** and **3D print**

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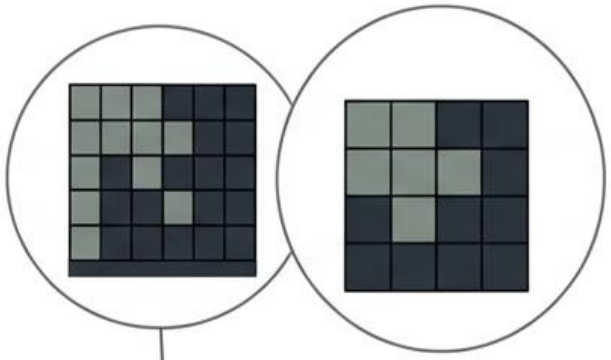
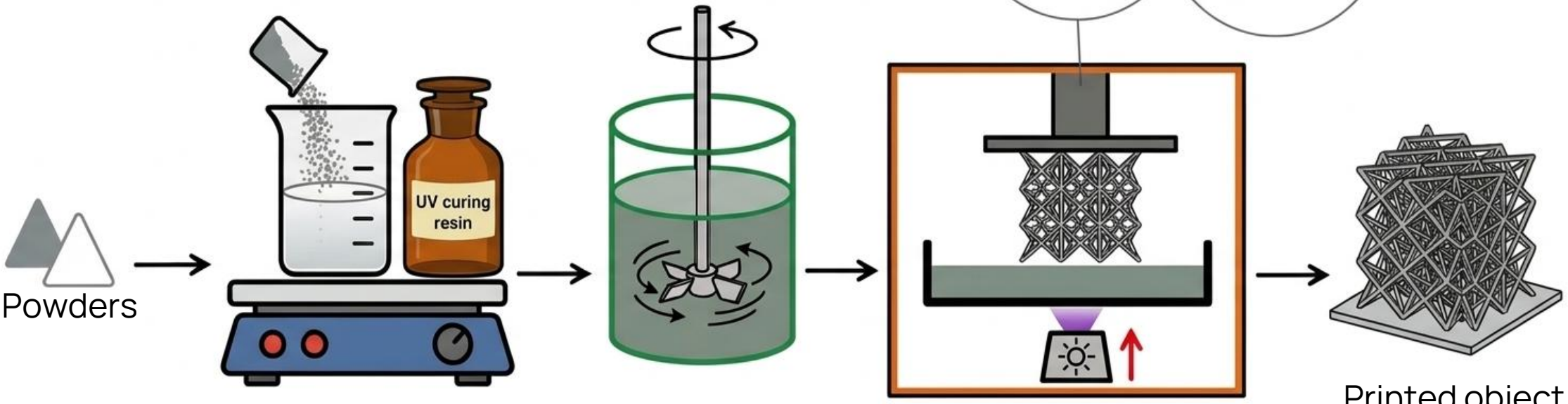
EIC THREAD

Slide credit: Jacopo Domaschio,
for details on particle loaded DLP, see Rontini et al., https://doi.org/10.1016/j.powtec.2025.122053

DLP CONCEPT: mix thermite with resin, which gives structural integrity and shaping capabilities

Each layer is cured according to the imposed light pattern





Powders → Add photocuring resin as binder → Mixing → DLP 3D printing → Printed object

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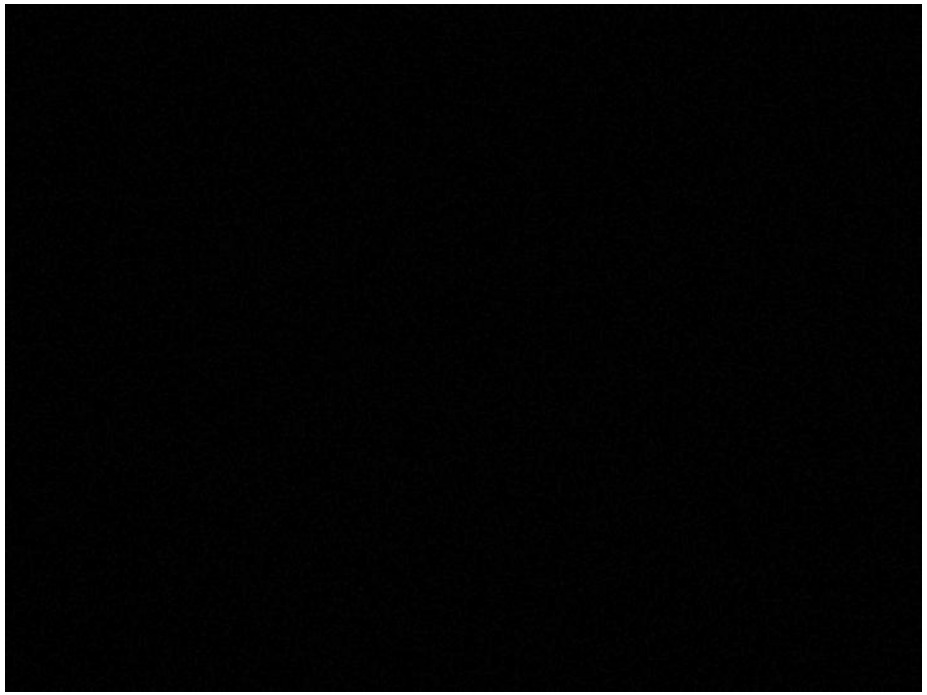
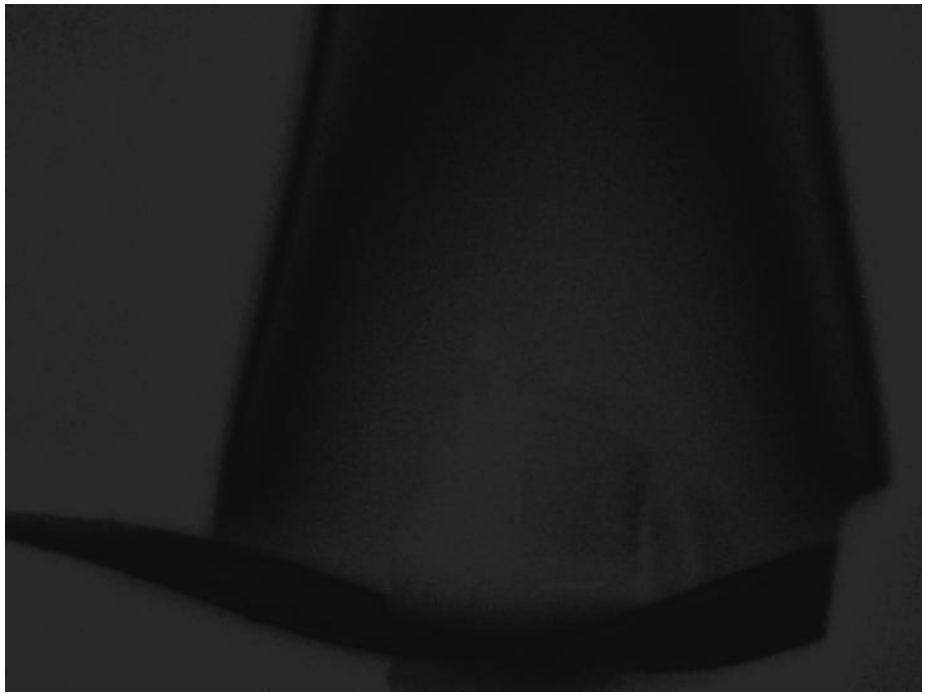
EIC THREAD

Slide credit: Jacopo Domaschio, Data Collection: Federica Catani, Valentina D'Annunzio, Jacopo Domaschio, Carlo Zanardi.



Sint. Al/WO_3: >200 mm/s



DLP Mg/SiO_2: ~17 mm/s

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EIC THREAD: Thermal Analysis – Tested Materials

Thermite:
- Mg - SiO_2 (loose powder) (Stoichiometric) in terms of composition
- Al-WO_3 (loose powder) (60-40 wt%)

Building Block (DLP):
- Mg - SiO_2 (60 wt%) in DLP resin (40 wt.%, cured) (Stoichiometric)

Material Reactivity Tailoring:
- Al-WO_3 (loose powder) – Activated (30 wt.%-70wt.%, and 40 wt.%-60wt.%)

Testing Conditions: for thermal analysis
- Heating rate from 2.5 to 20 K/min
- Ar flow (50 ml/min)
- 0.1 MPa
- Reference pan: alpha-alumina

Notes:
- Aluminum is micron-sized (30 um nominal size, active metal content > 98 wt.%)
- Magnesium is micron-sized (44 um nominal size)
- Oxides are micron-sized

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A1: Requirement

- Produce an analysis providing relevant parameters for kinetic modelling of the thermite (say, Arrhenius parameters, isoconversional analysis).
- Contrast the effects of mechanical mixing with mechanical activation (loose thermite powder, simply mixed vs. thermite powder with metal/metal oxide combined by mechanical activation.
- Evaluate the impact of the DLP printing on the thermite reaction.

starting point: from rough data produce graphs using time and temperature data.

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